

Southeast University

Department of Computer Science and Engineering
Program: B.Sc. Eng. in CSE
Course Code: CSE346.2
Course Title: ISD & SE Lab

PROJECT REPORT

Employee Management System

Farhan Ishraque
ID: 2023000000042

Sanjid Islam Joy
ID: 2022000000144

Most. Jannat Binta Mahmud
ID: 2023000000144

Submitted To
Shimul Dey Katha
Lecturer
Department of Computer Science & Engineering
Southeast University

Employee Management System

Goal:

To design and implement an efficient, user-friendly secure system that enables organizations to manage employee information, track attendance and leaves, organize departments and provide HR-related tasks through a centralized platform.

Feature Identification:

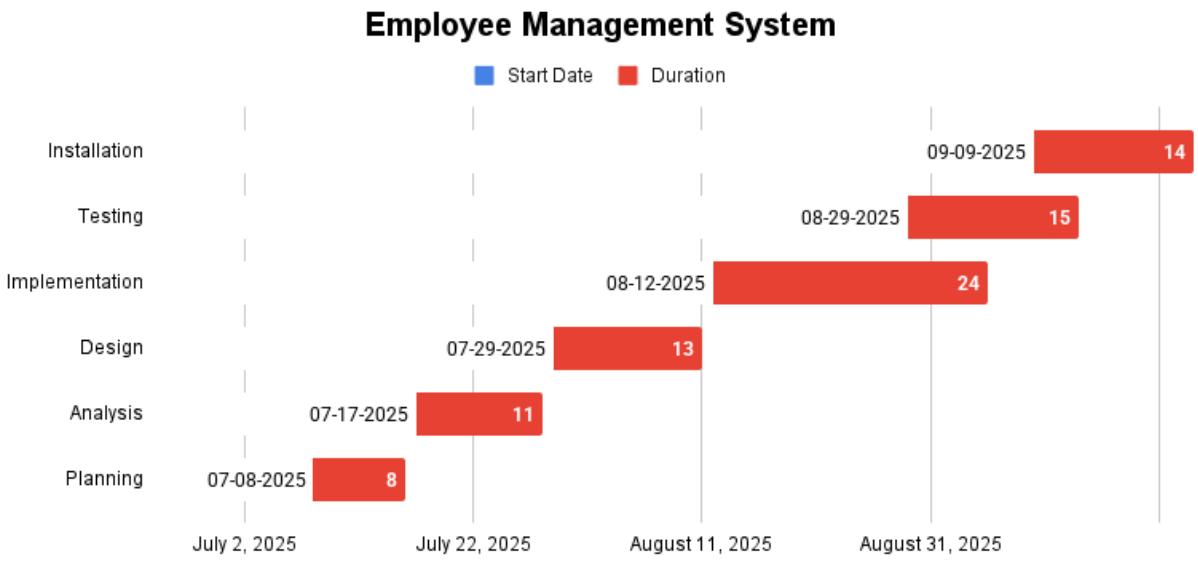
Employee Panel:

1. Log in
2. View profile
3. Attendance view
4. Leave Management
5. Announcement

HR Panel:

1. Dashboard overview
2. Manage employee records
3. Role and access control
4. Department management
5. General reports(Attendance, leave history, department wise employee list)

Gantt Chart:


Employee Management System
Start Date
Duration
Installation
Testing
Implementation
Design
Analysis
Planning
09-09-2025
14
08-29-2025
15
08-12-2025
24
07-29-2025
13
07-17-2025
11
07-08-2025
8
July 2, 2025
July 22, 2025
August 11, 2025
August 31, 2025

Budget Details:

SL	Criteria	Cost Specification	Cost(BDT)
1	Office Cost	Office Rent	60,000
		Utilities	10,000
		Cleaning and Maintenance	5,000
		Team Meeting	20,000
		Project Meeting	20,000
2	Website Cost	Domain + Hosting	5,000
		SSL Certificate	1,000
		Maintenance and Update	10,000
3	Office Equipment	Computer	5,00,000
		Laptop	2,00,000
		Furniture	7,000
4	Salary Cost	Project Manager	1,50,000
		Team Leader	1,00,000
		Developer(2 person)	1,20,000
		UI/UX Designer	50,000
		Test Engineer	40,000
		Admin	30,000
	Total cost		**13,28,000**

Requirement Analysis

1. System Requirements:

Hardware Requirements:

Server

- Minimum 8 GB RAM
- 500 GB SSD storage
- Quad-core processor

Client Devices

- Stable internet connection (required for Figma’s cloud service)
- Minimum screen resolution: 1920 x 1080 pixels

Software Requirements:

- Design Tool: Figma (web-based UI/UX design and prototyping tool)
- Database: MySQL
- Operating System: Windows, macOS, Linux (via web browser)
- Browsers: Latest versions of Google Chrome, Mozilla Firefox, Microsoft Edge, Safari

2.User Requirements:

For HR/Admin

- Secure login and access control
- Add, edit and delete employee records
- Assign departments, roles and reporting managers
- Generate reports (attendance, leaves, department-wise etc.)
- Set performance or attendance thresholds

For Employees

- View personal attendance and leave records
- Apply for leave
- Receive notifications (low attendance, leave approval)

3.Functional Requirements

Employee Panel

- **FR1:** The system shall allow employees to log in securely with unique credentials.
- **FR2:** The system shall allow employees to view and update their profile information.
- **FR3:** The system shall allow employees to view their daily and monthly attendance records.
- **FR4:** The system shall allow employees to apply for and manage leave requests.
- **FR5:** The system shall display company announcements to employees in real-time.

HR Panel

- **FR6:** The system shall provide HR with a dashboard overview of attendance, leave and employee statistics.
- **FR7:** The system shall allow HR to create, edit and delete employee records.
- **FR8:** The system shall enable HR to assign roles and manage access control for different user types.
- **FR9:** The system shall allow HR to manage departments and assign employees accordingly.
- **FR10:** The system shall generate general reports such as attendance summaries, leave history and department-wise employee lists.

4.Non-Functional Requirements

- **Performance:** The system should load pages and dashboard data within 3 seconds.
- **Scalability:** The system must support up to 2,000 concurrent users including employees and HR staff.
- **Security:** All user credentials and sensitive data must be encrypted. Role-based access control must be enforced.
- **Usability:** The user interface should be simple, intuitive and fully responsive on desktops, tablets and smartphones.
- **Reliability:** The system uptime must be at least 99% annually with minimal downtime for maintenance.
- **Maintainability:** The codebase should be modular, well- documented and follow best coding practices to ease future updates
- **Backup:** Daily automatic backups of the database and system configurations must be scheduled and stored securely.

SDLC Model Selection:

Criteria and Priority Weights:

Criteria	Weight(Priority)	Reason
Requirement Flexibility	4	HR policies and company structures may change, requiring system adaptability.
Customer Involvement	4	Active HR and management participation is needed for accurate requirements.
Project Size	2	The system is small scale, typically within a single organization.
Risk Handling	3	Medium risk; mainly operational rather than financial.
Testing Focus	4	Data accuracy and functionality are essential to avoid payroll/attendance errors.
Security Requirement	5	Sensitive employee personal and salary data must be protected.
Cost of Change	4	Frequent policy changes mean updates must be affordable.
Delivery Speed	3	Moderate speed is important to deploy system improvements quickly.
User-Friendliness (UI/UX)	5	All employees may interact with the system; ease of use is critical.
Scalability	3	Must handle future company growth without major redesign.

Rate Each Model per Criterion:

Criteria	Weight (Priority)	Waterfall	V	Iterative	Incremental	Spiral	Agile
Requirement Flexibility	4	1	1	4	4	5	5
Customer Involvement	4	1	1	3	3	4	5
Project Size	2	3	3	4	4	5	4
Risk Handling	3	2	2	3	3	5	4
Testing Focus	4	3	5	4	4	5	5
Security Requirement	5	3	5	3	4	5	4
Cost of Change	4	1	1	3	3	5	5
Delivery Speed	3	2	2	3	4	3	5
User Friendliness (UI/UX)	5	2	2	4	4	5	5
Scalability	3	2	2	4	4	5	4

Calculation of Weighted Scores:

Formula: Weighted Score = Weight × Rating

Criteria	Weight (Priority)	Waterfall	V	Iterative	Incremental	Spiral	Agile
Requirement Flexibility	4	4	4	16	16	20	20
Customer Involvement	4	4	4	12	12	16	20
Project Size	2	6	6	8	8	10	8
Risk Handling	3	6	6	9	9	15	12
Testing Focus	4	12	20	16	16	20	20
Security Requirement	5	15	25	15	20	25	20
Cost of Change	4	4	4	12	12	20	20
Delivery Speed	3	6	6	9	12	9	15
User Friendliness (UI/UX)	5	10	10	20	20	25	25
Scalability	3	6	6	12	12	15	12
Total		**73**	**91**	**129**	**137**	**175**	**172**

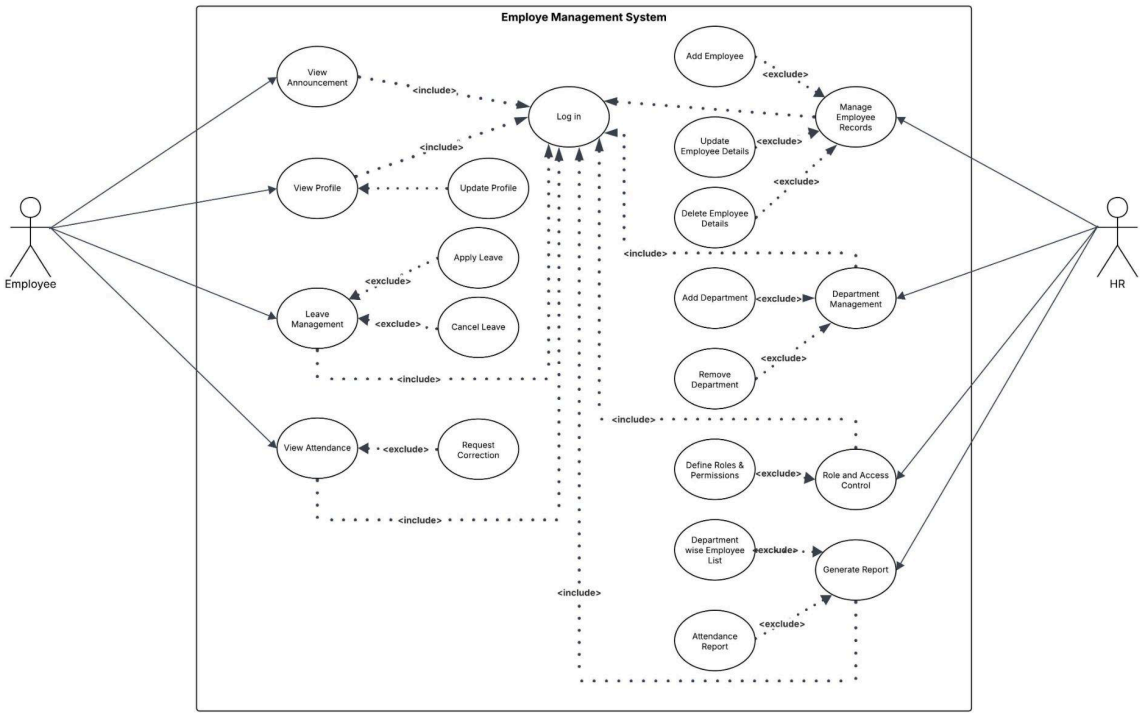
Justification:

The **Spiral Model** scores the highest (**175**) for EMS development, making it the most suitable choice due to its strong focus on flexibility, security, and iterative risk management. Agile (172) is a close second and could be used for modules requiring rapid updates, such as leave tracking or employee feedback systems.

Use Case Diagram:

Actors: Employee, HR

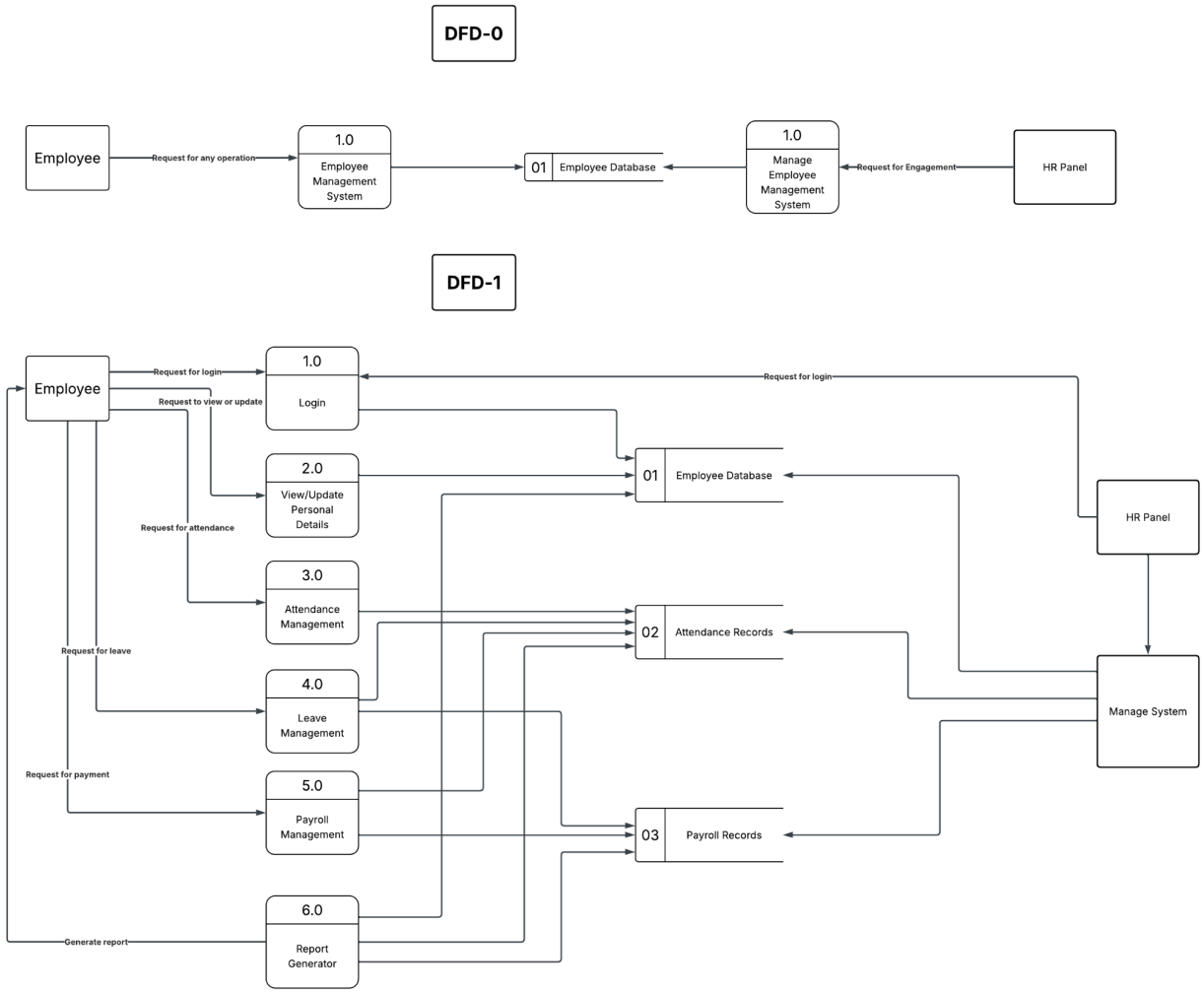
- **Employee:** Login, View Profile/Attendance, Manage Leaves, View Announcements.
- **HR/Admin:** Manage all records, control access & roles, manage departments, generate reports.



Data Flow Diagram:

Illustrates how data moves between external entities, processes and data stores.

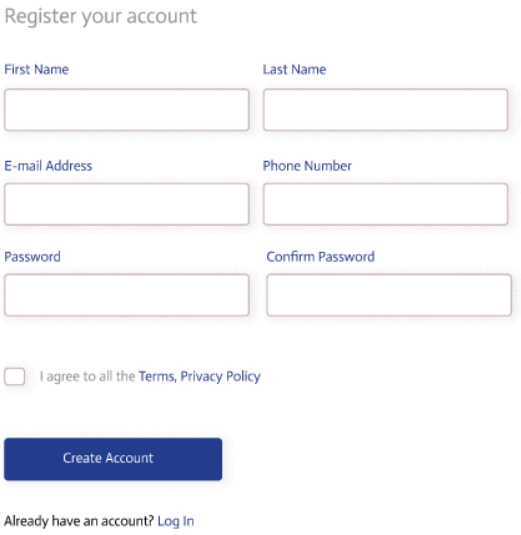
- **Entities:** Employee, HR
- **Processes:** Login Authentication, Attendance Management, Leave Management, Payroll Management, Report Generation
- **Data Stores:** Employee Database, Attendance Records, Payroll Records



Implementation:

User Section:





Login

Login to your account.

E-mail Address

Password

Remember me

Reset Password?

Sign In

Don't have an account yet? Sign Up



Dashboard Payroll Company Extras

Dashboard



Farhan Ishraque

Employee

Edit Profile

Quick Actions

Update Profile | Apply Leave | Take Appraisal | View Payslip

Available Leave Days

Annual Leave	10 of 60 day(s)
Sick Leave	0 of 10 day(s)
Compassionate Leave	8 of 15 day(s)

To-dos

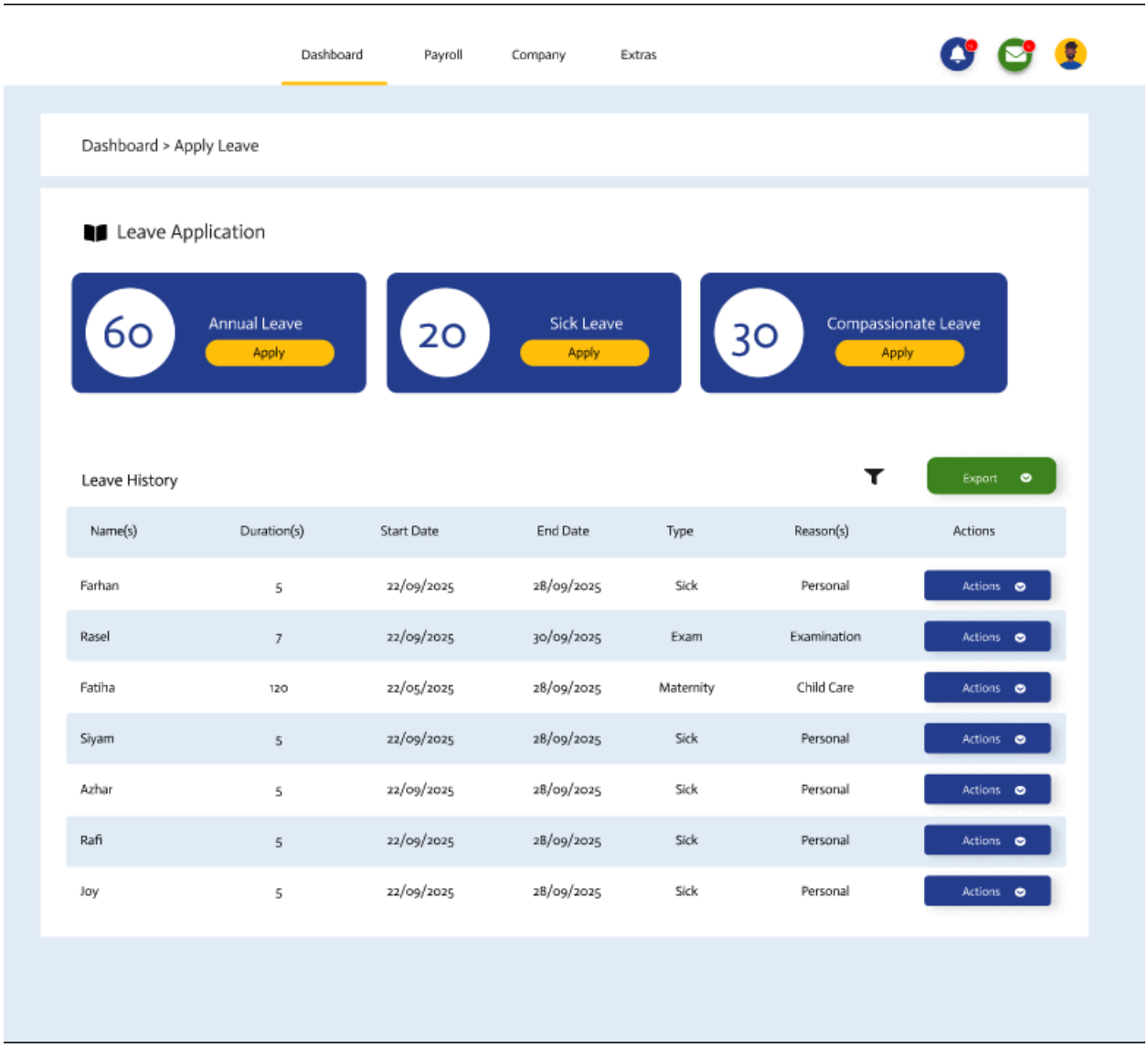
- Complete document upload
- Follow up on client documents
- Design wire frames for IUB project
- Create case study for next IT project
- Follow up for next project

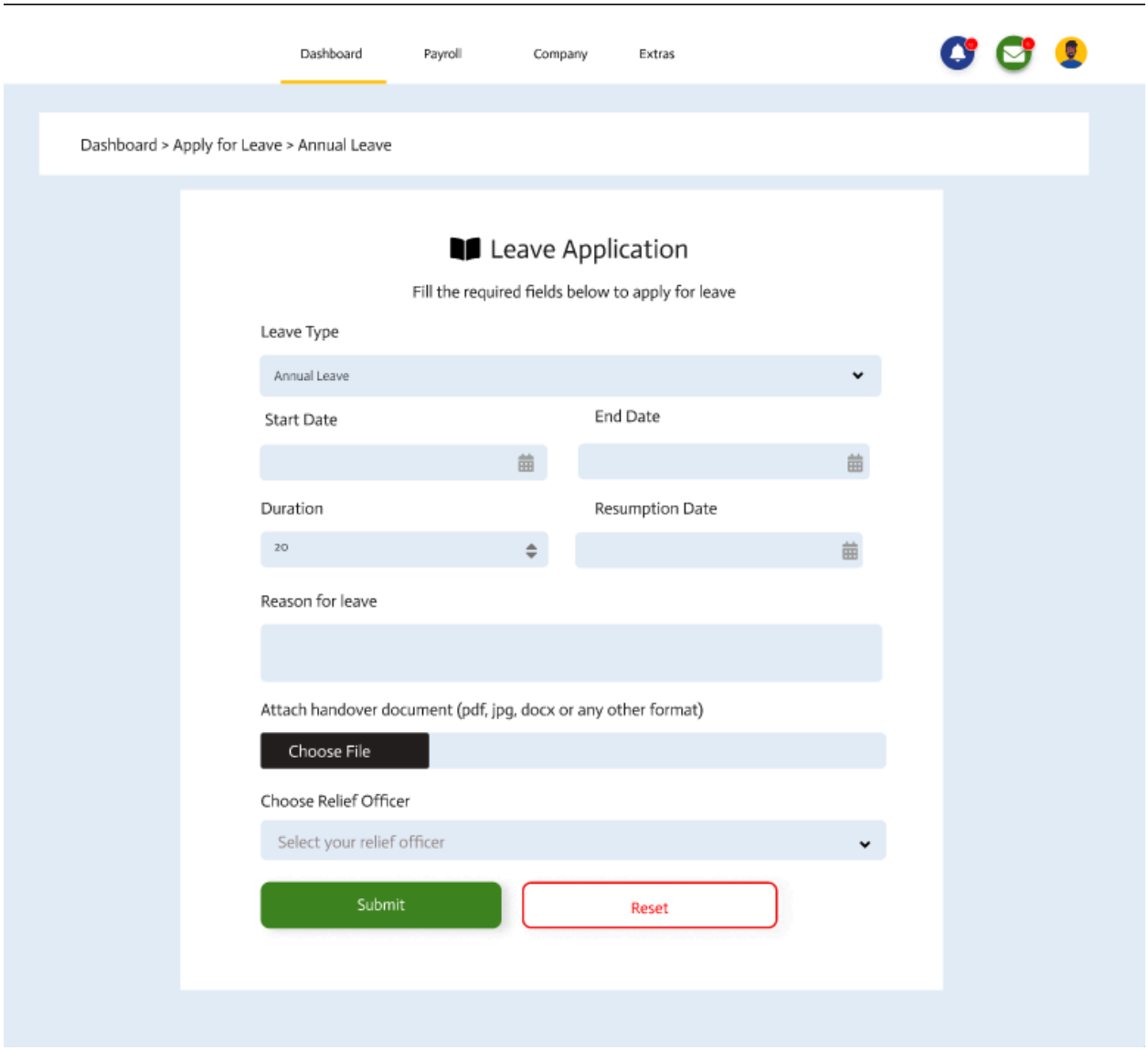
Announcements

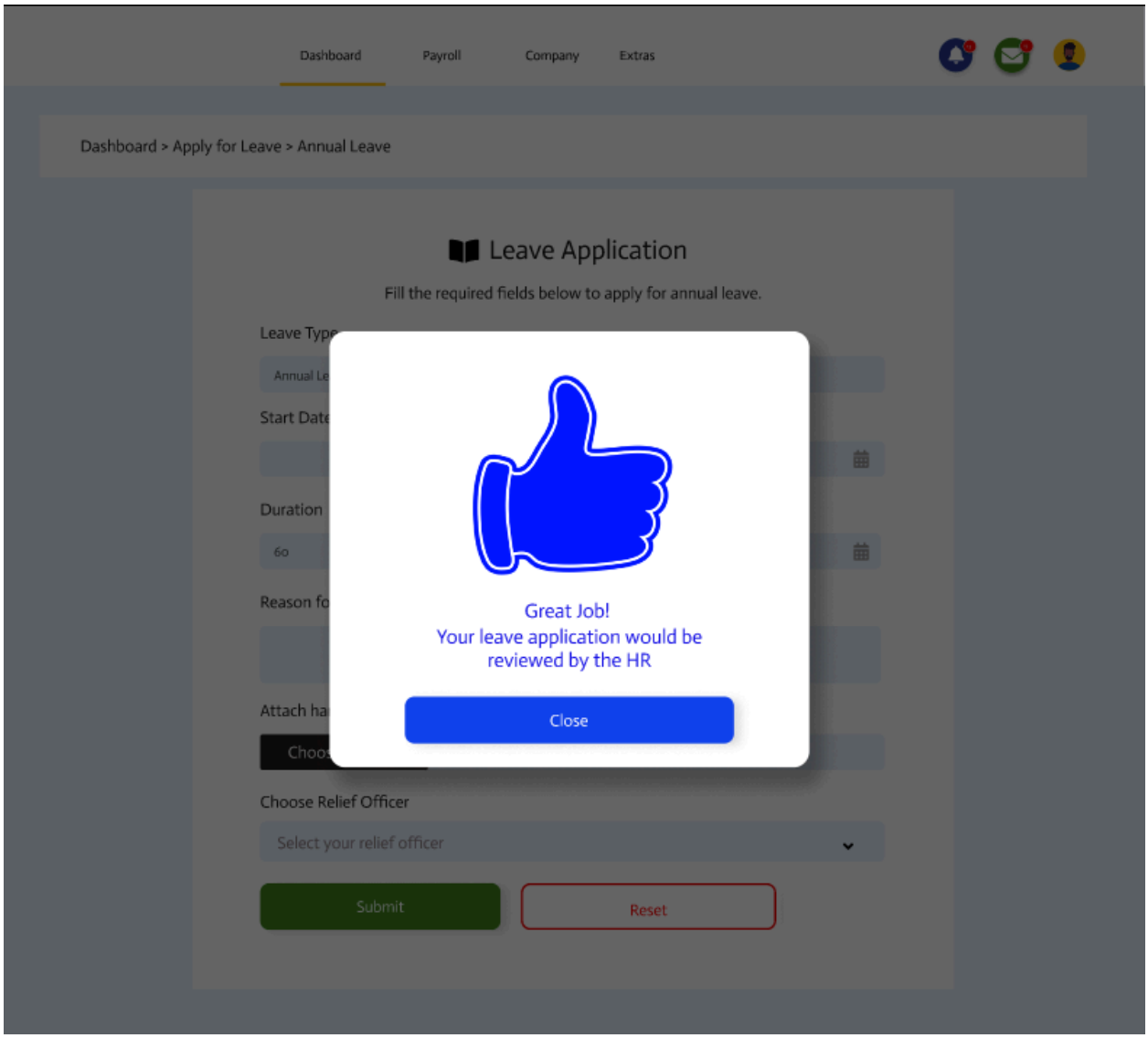
- Welcome Joy - We have a new staff joining us
- Meeting for new project : Kindly gather at the meeting hall
- Marriage Alert
- Office Space Update

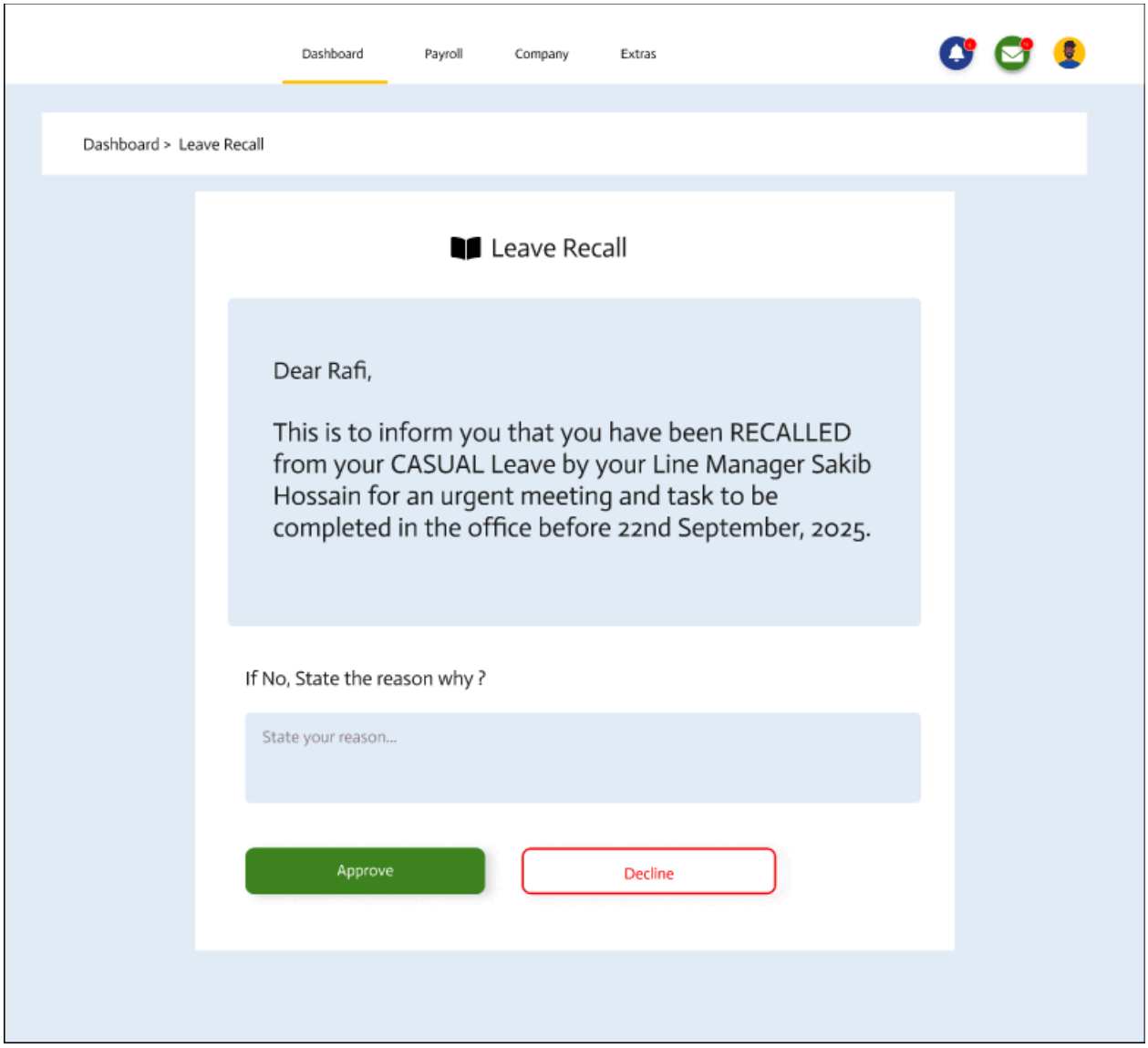
September Pay Slip Breakdown

Earnings	Amount	Deductions	Total
Basic Wage	150,000	-30,000	120,000
Tax	15,000	-3,000	12,000
Total Earnings	165,000	-33,000	132,000

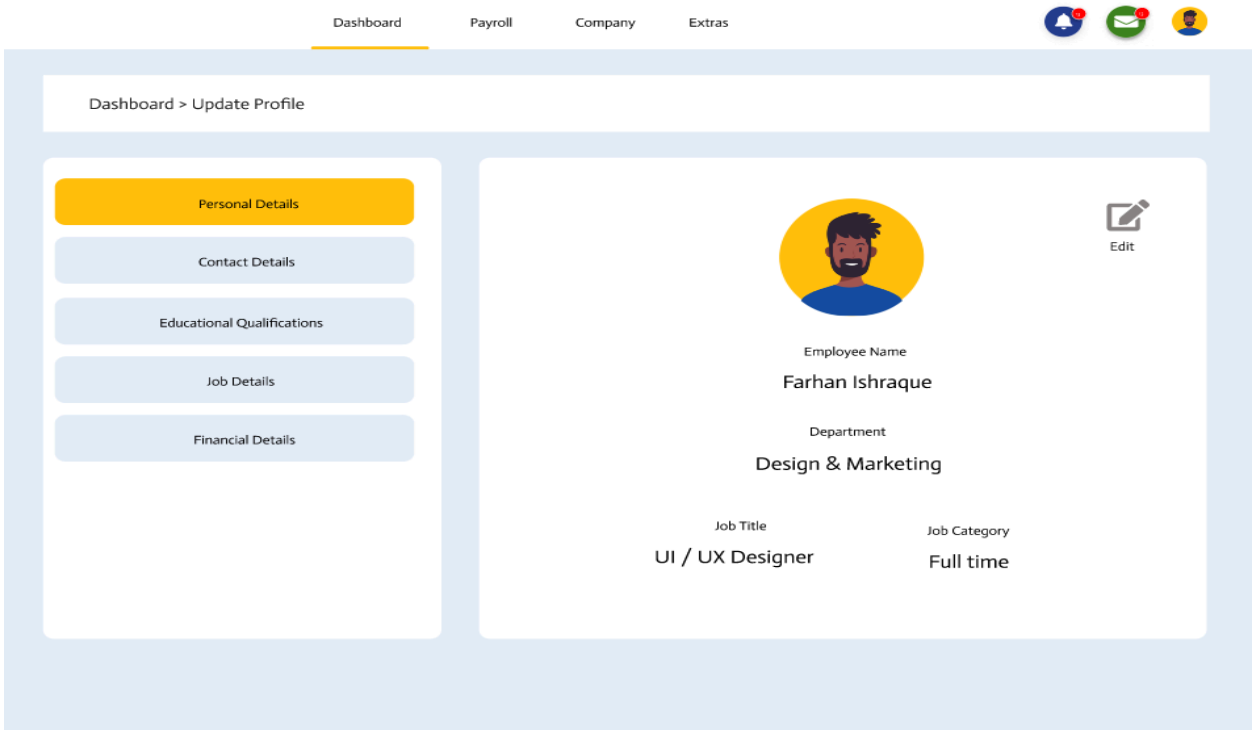

Dashboard
Payroll
Company
Extras
Dashboard > Apply Leave
Leave Application
60
Annual Leave
Apply
20
Sick Leave
Apply
30
Compassionate Leave
Apply
Leave History
Export
Name(s) Duration(s) Start Date End Date Type Reason(s) Actions
Farhan 5 22/09/2025 28/09/2025 Sick Personal Actions
Rasel 7 22/09/2025 30/09/2025 Exam Examination Actions
Fatiha 120 22/05/2025 28/09/2025 Maternity Child Care Actions
Siyam 5 22/09/2025 28/09/2025 Sick Personal Actions
Azhar 5 22/09/2025 28/09/2025 Sick Personal Actions
Rafi 5 22/09/2025 28/09/2025 Sick Personal Actions
Joy 5 22/09/2025 28/09/2025 Sick Personal Actions

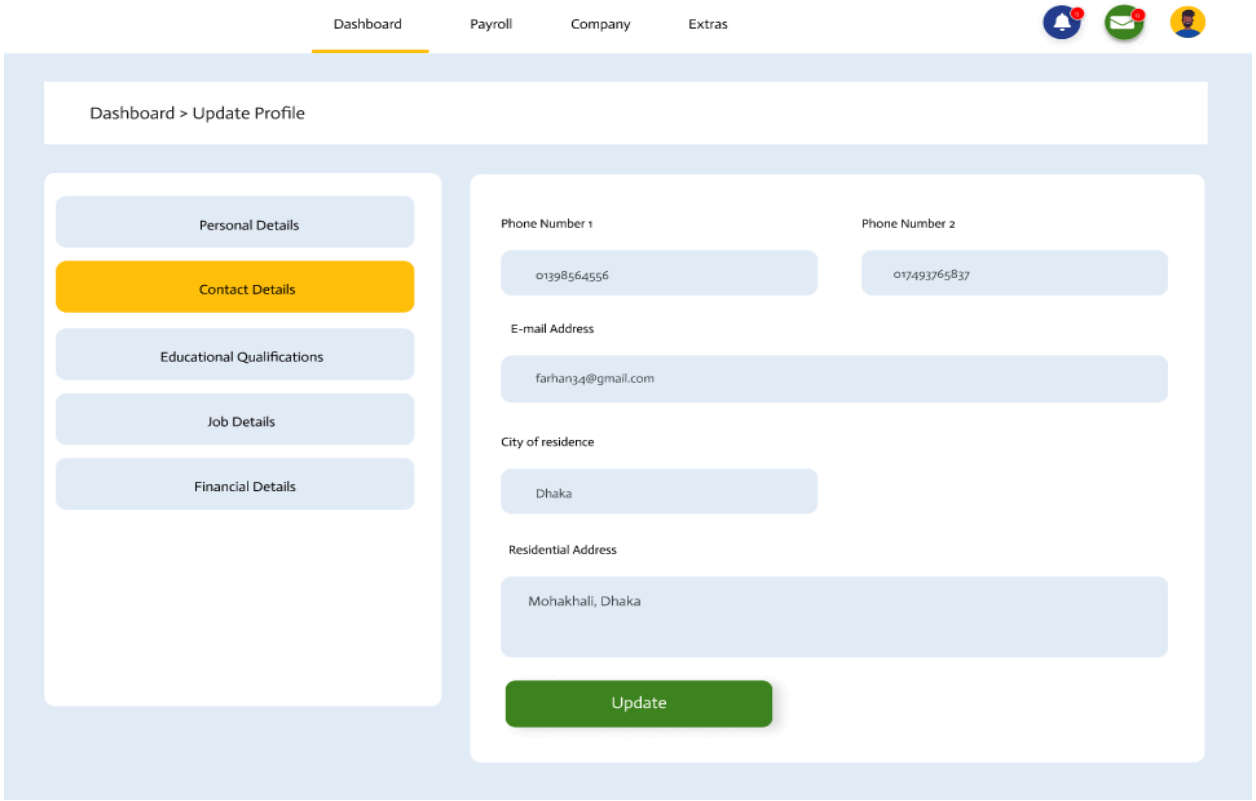

Dashboard
Payroll
Company
Extras
Dashboard > Apply for Leave > Annual Leave
Leave Application
Fill the required fields below to apply for leave
Leave Type
Annual Leave
Start Date
End Date
Duration
20
Resumption Date
Reason for leave
Attach handover document (pdf, jpg, docx or any other format)
Choose File
Choose Relief Officer
Select your relief officer
Submit
Reset

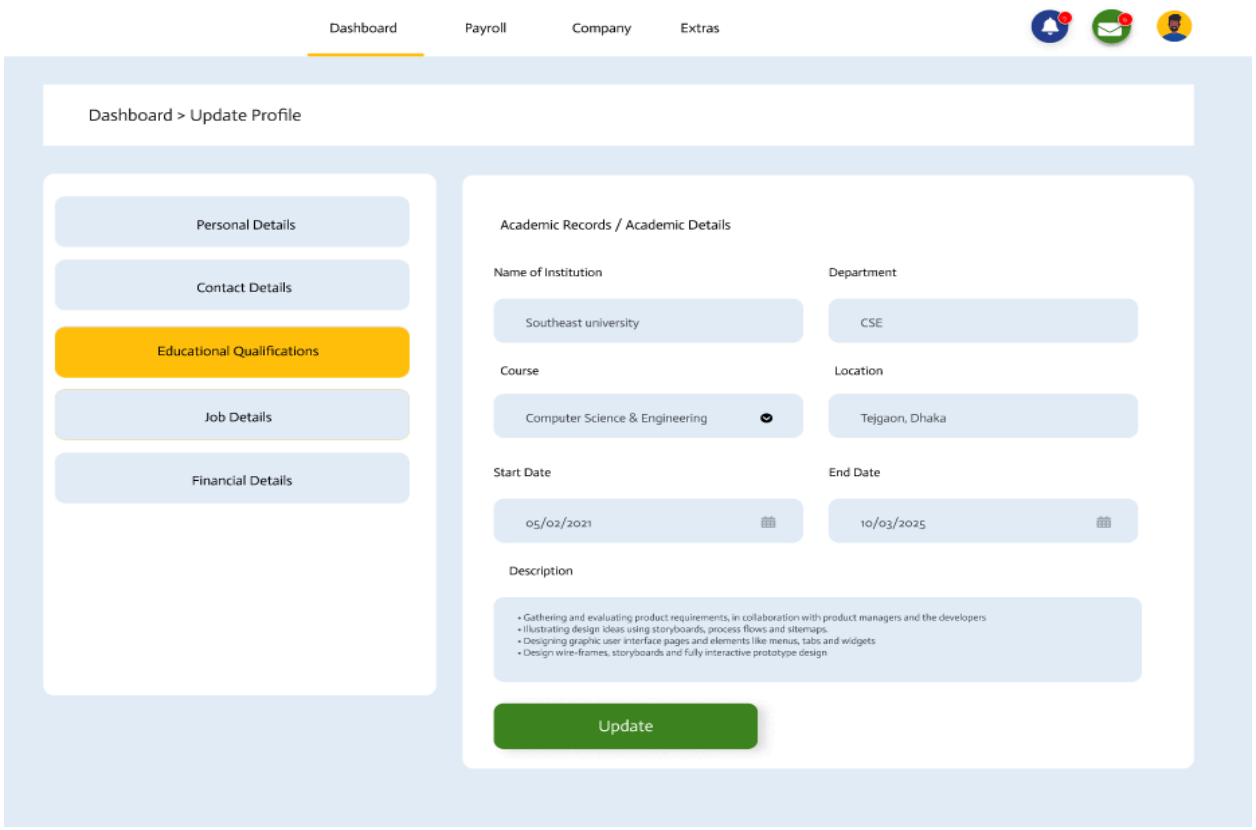

Dashboard
Payroll
Company
Extras
Dashboard > Apply for Leave > Annual Leave
Leave Application
Fill the required fields below to apply for annual leave.
Leave Type
Annual Le
Start Date
Duration
60
Reason fo
Attach ha
Choos
Choose Relief Officer
Select your relief officer
Submit
Reset
Great Job!
Your leave application would be reviewed by the HR
Close

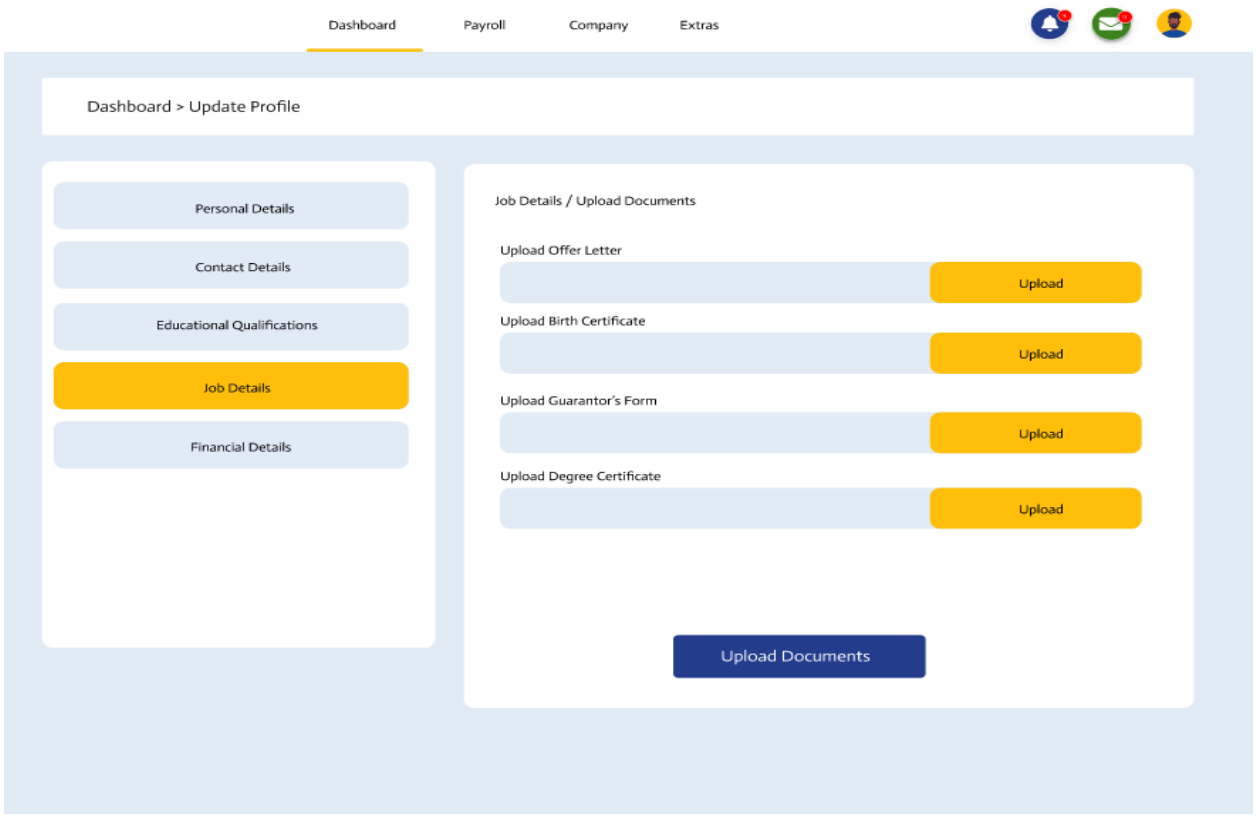

Dashboard
Payroll
Company
Extras
Dashboard > Leave Recall
Leave Recall
Dear Rafi,
This is to inform you that you have been RECALLED from your CASUAL Leave by your Line Manager Sakib Hossain for an urgent meeting and task to be completed in the office before 22nd September, 2025.
If No, State the reason why ?
State your reason...
Approve
Decline

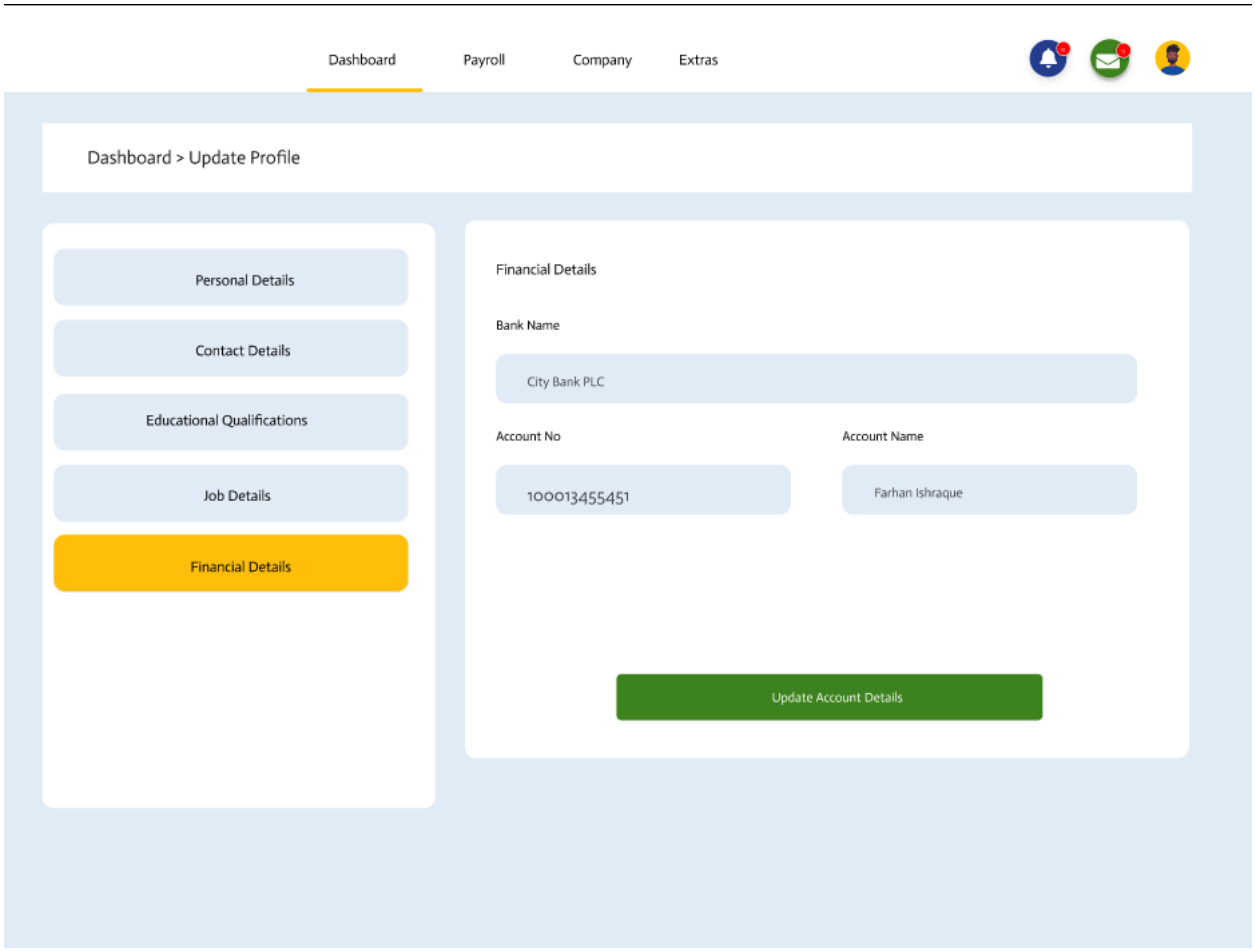
Update Profile:



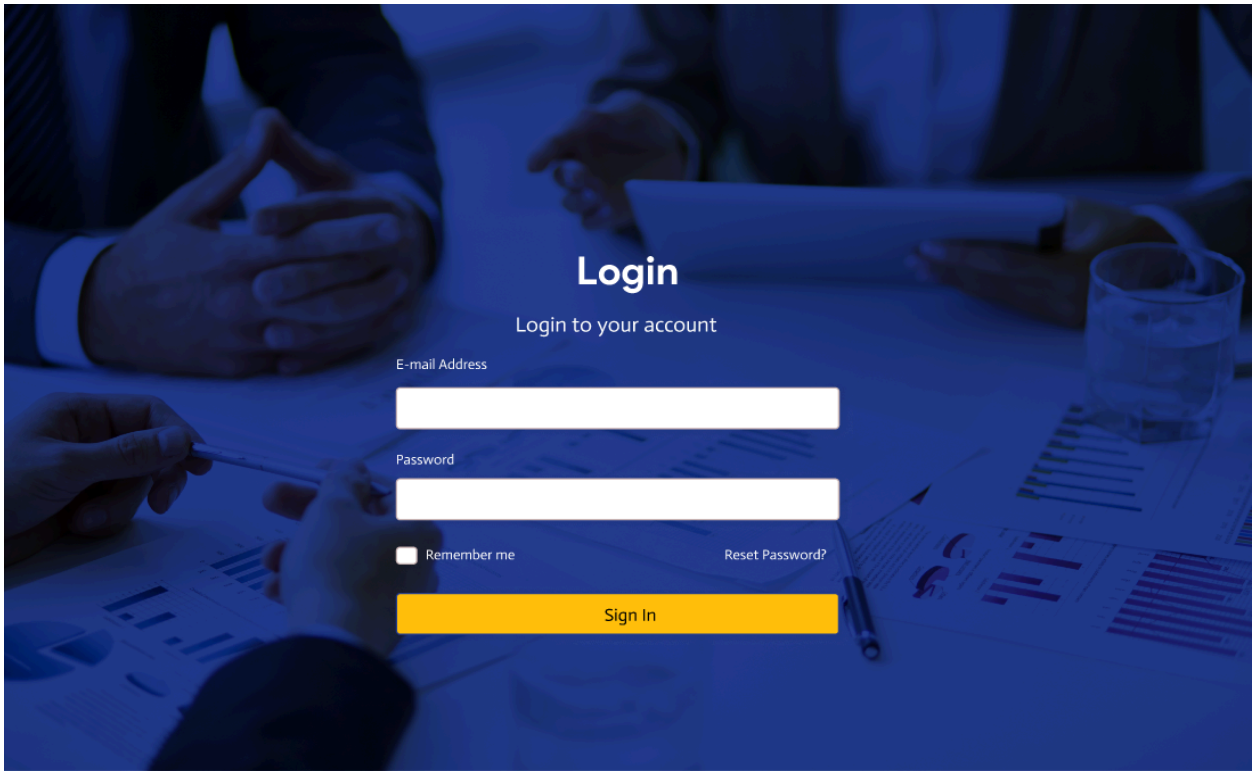


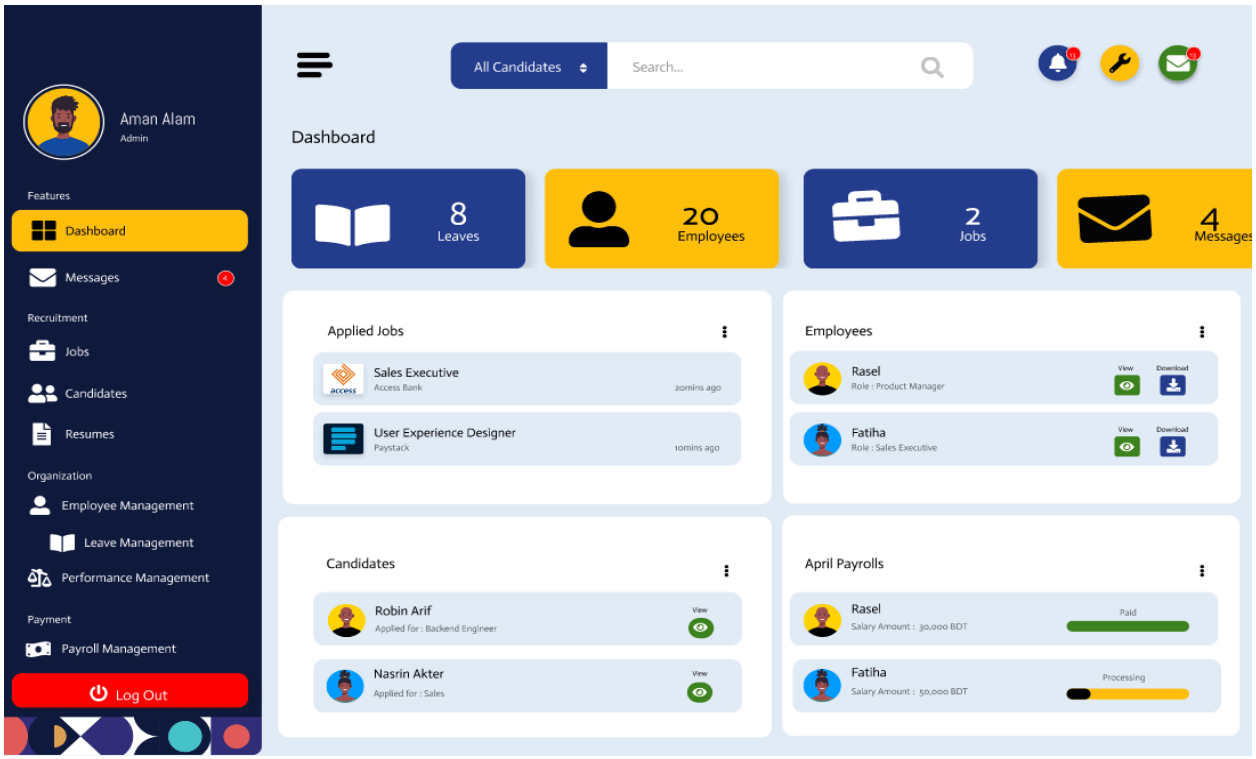

Dashboard
Payroll
Company
Extras
Dashboard > Update Profile
Personal Details
Contact Details
Educational Qualifications
Job Details
Financial Details
Academic Records / Academic Details
Name of Institution
Southeast university
Department
CSE
Course
Computer Science & Engineering
Location
Tejgaon, Dhaka
Start Date
05/02/2021
End Date
10/03/2025
Description
• Gathering and evaluating product requirements, in collaboration with product managers and the developers
• Illustrating design ideas using storyboards, process flows and sitemaps.
• Designing graphic user interface pages and elements like menus, tabs and widgets
• Design wire-frames, storyboards and fully interactive prototype design
Update


Dashboard
Payroll
Company
Extras
Dashboard > Update Profile
Personal Details
Contact Details
Educational Qualifications
Job Details
Financial Details
Job Details / Upload Documents
Upload Offer Letter
Upload
Upload Birth Certificate
Upload
Upload Guarantor's Form
Upload
Upload Degree Certificate
Upload
Upload Documents

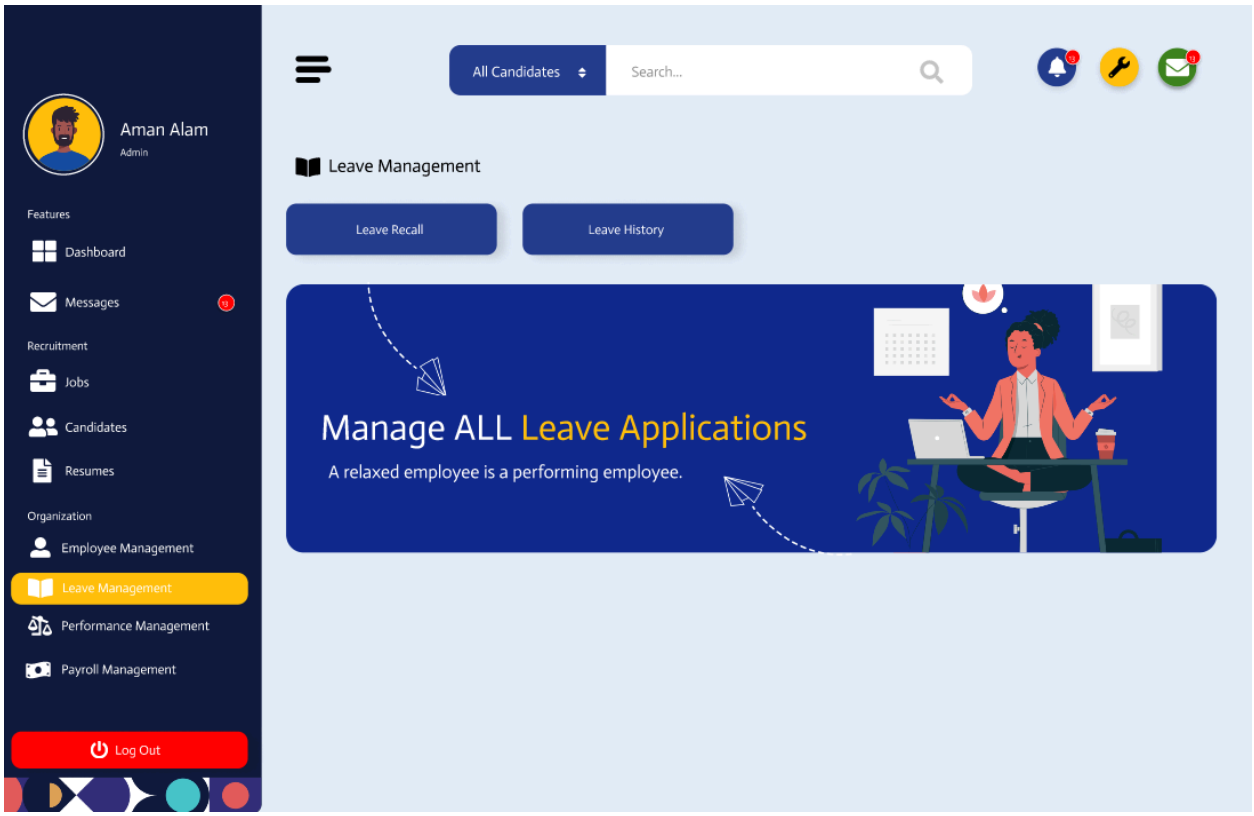

Dashboard
Payroll
Company
Extras
Dashboard > Update Profile
Personal Details
Contact Details
Educational Qualifications
Job Details
Financial Details
Financial Details
Bank Name
City Bank PLC
Account No
100013455451
Account Name
Farhan Ishraque
Update Account Details

Admin Section:

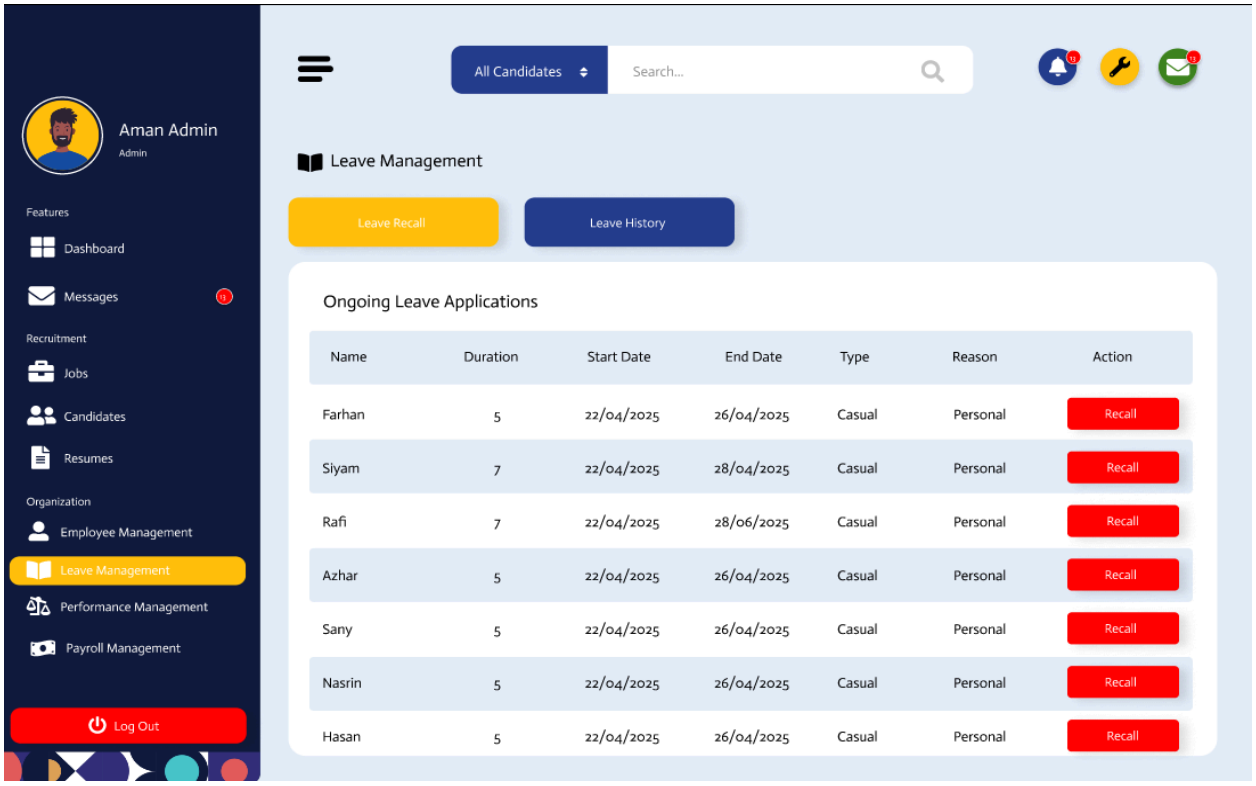


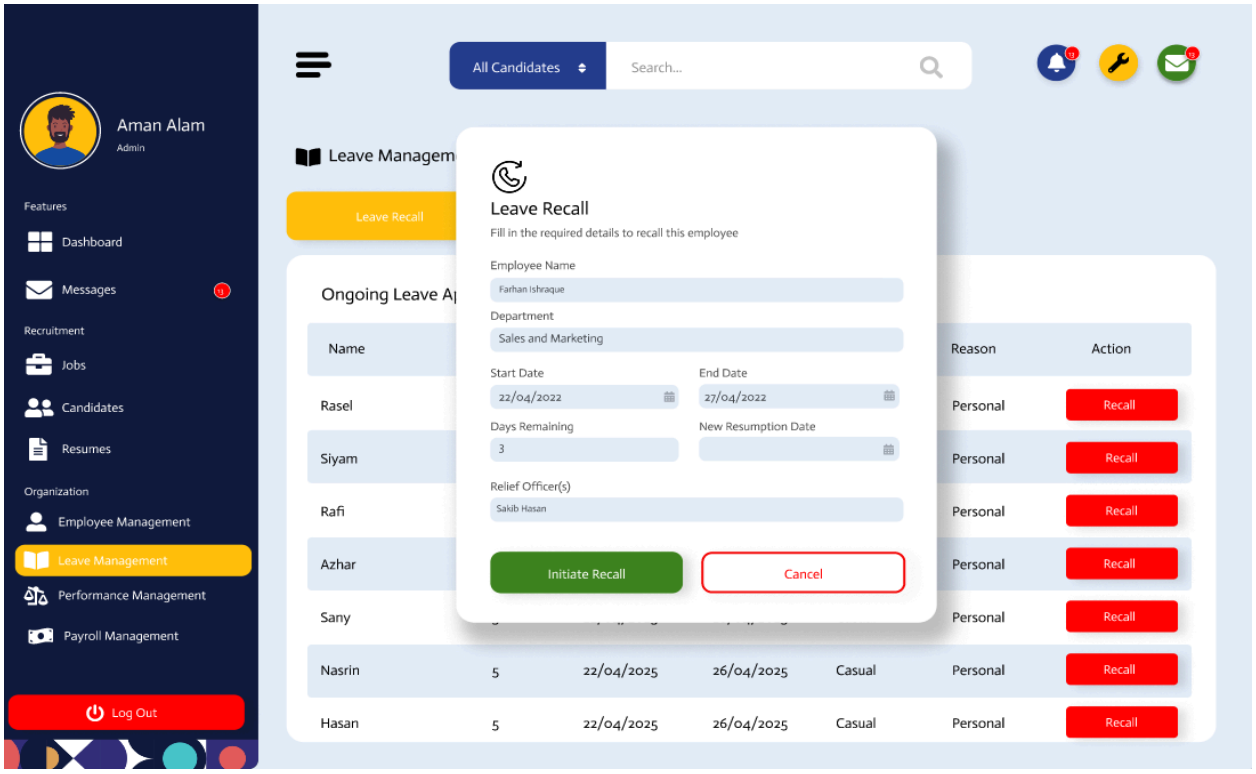


Leave Management

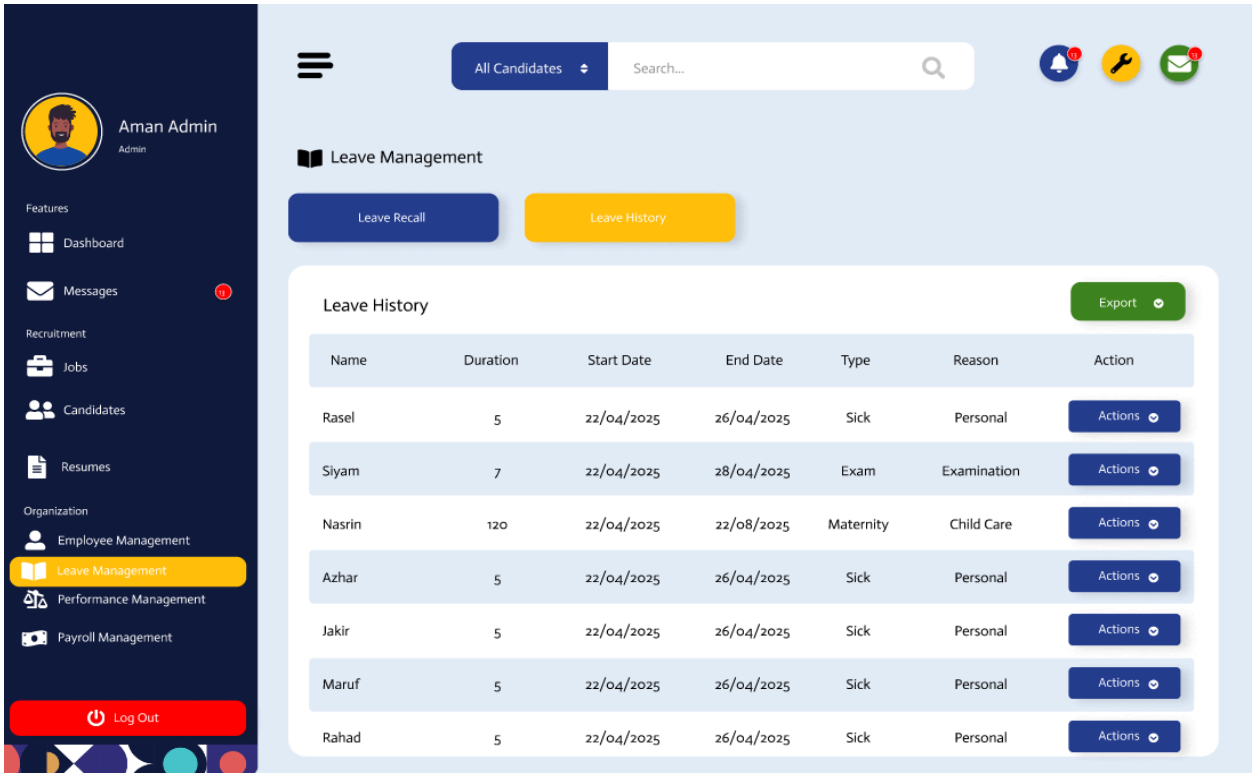


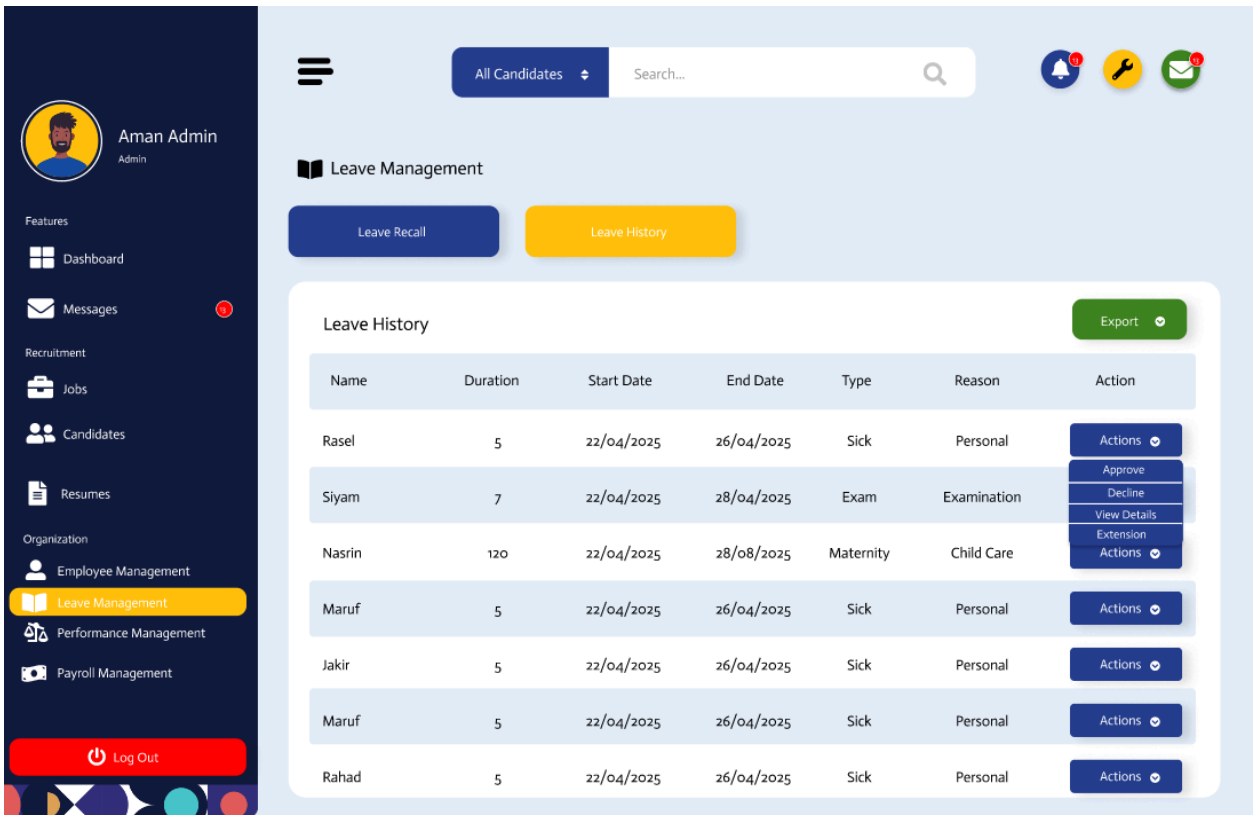
Leave Recall





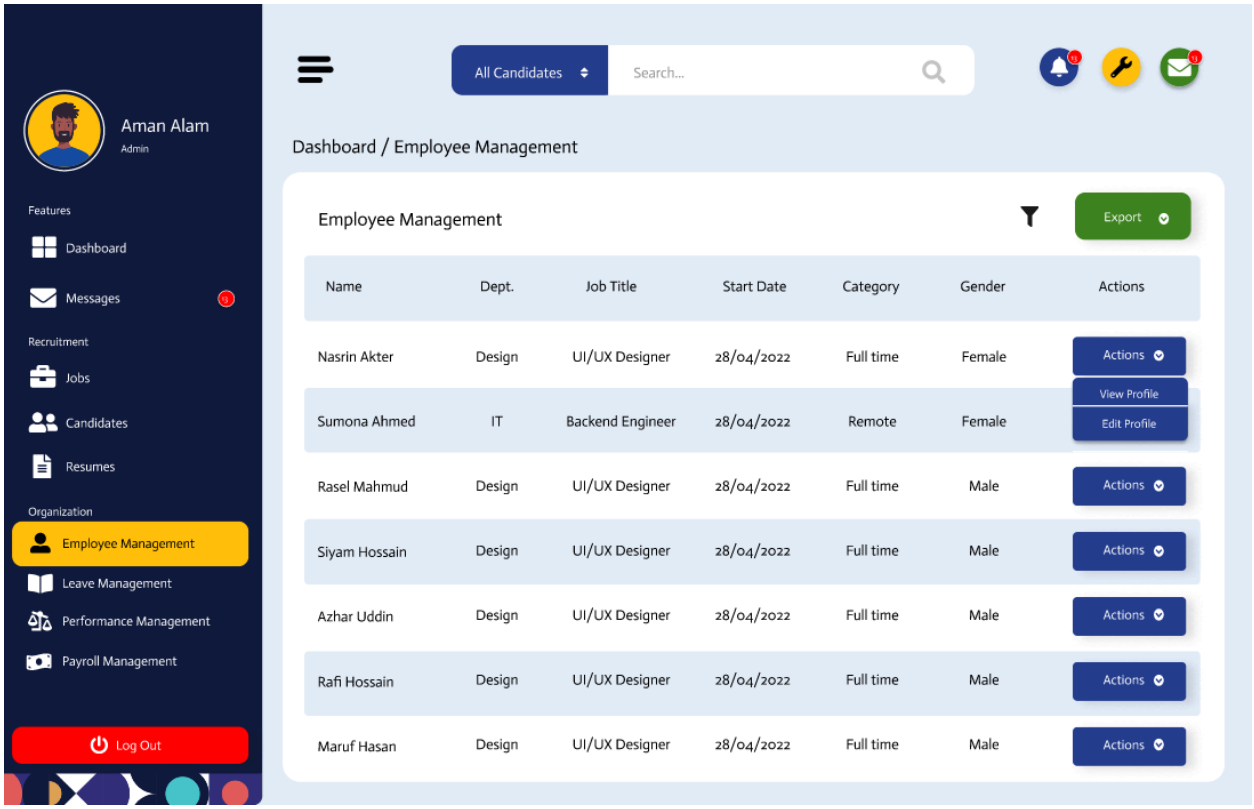
Leave History

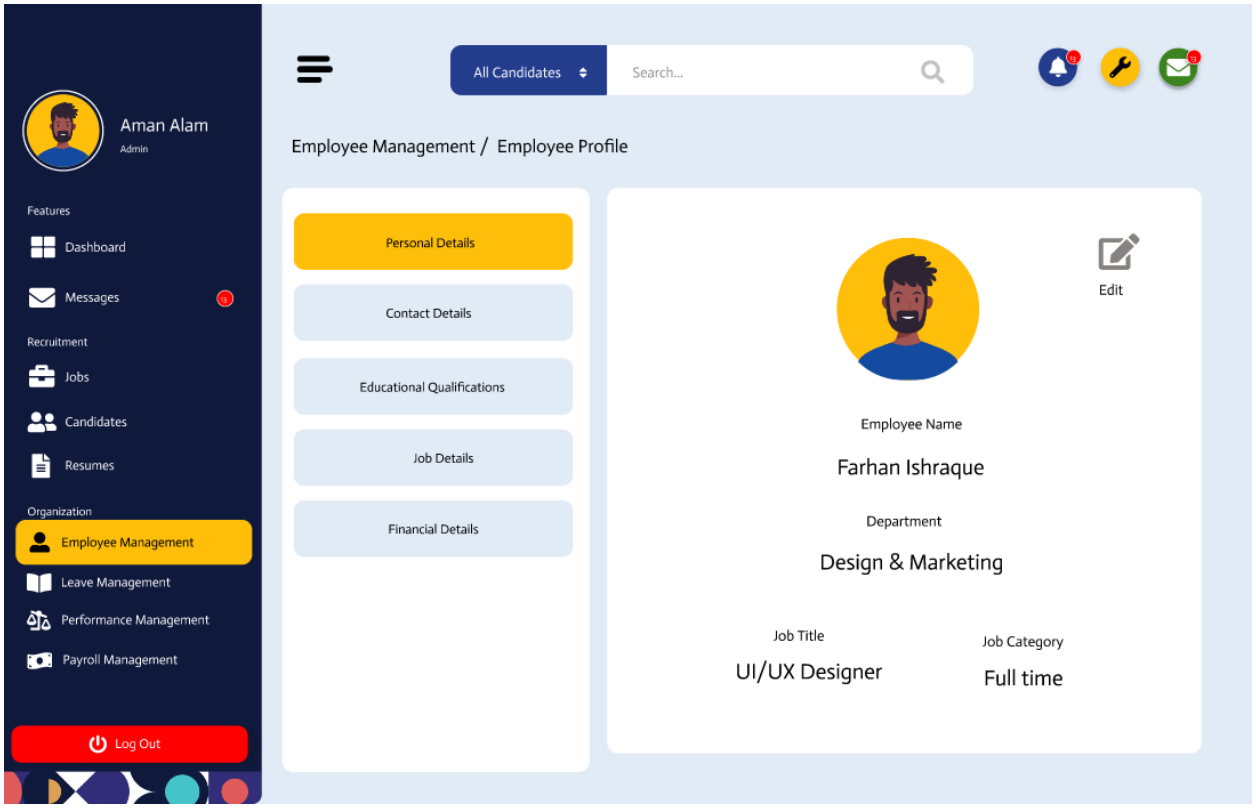


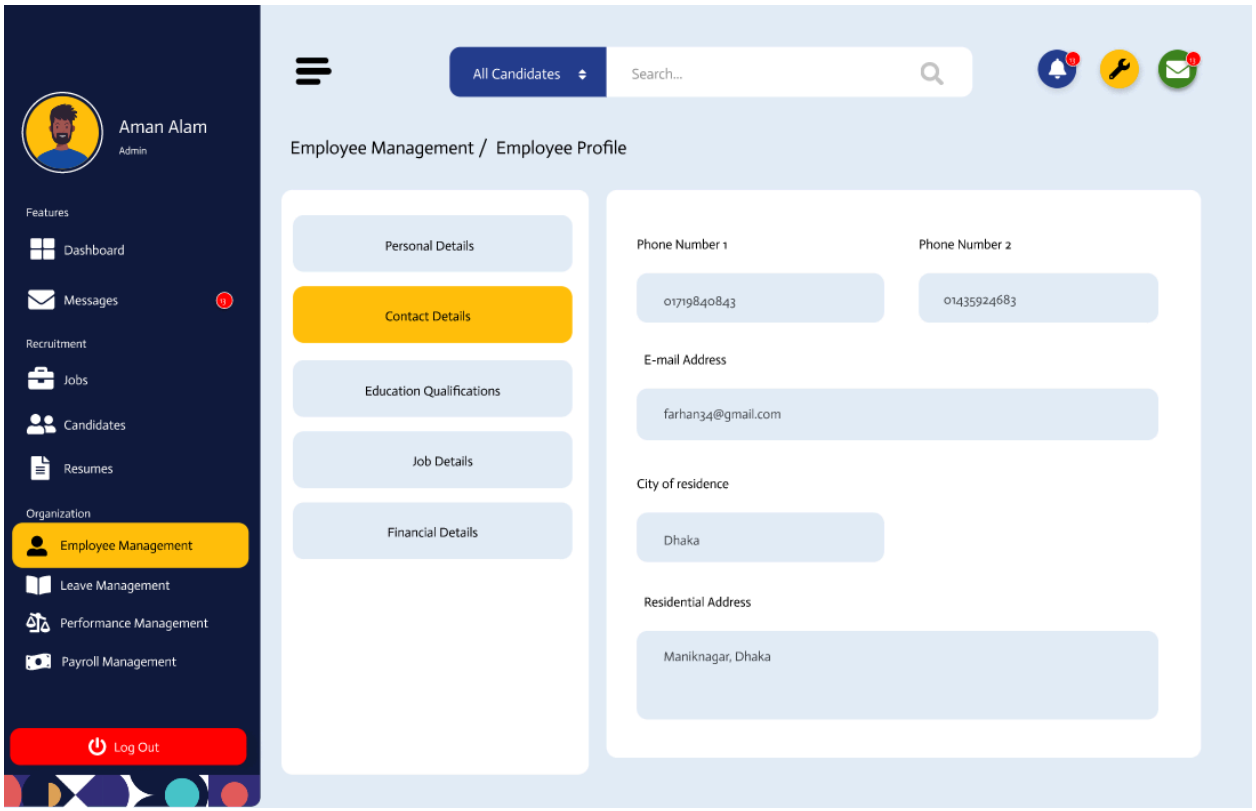


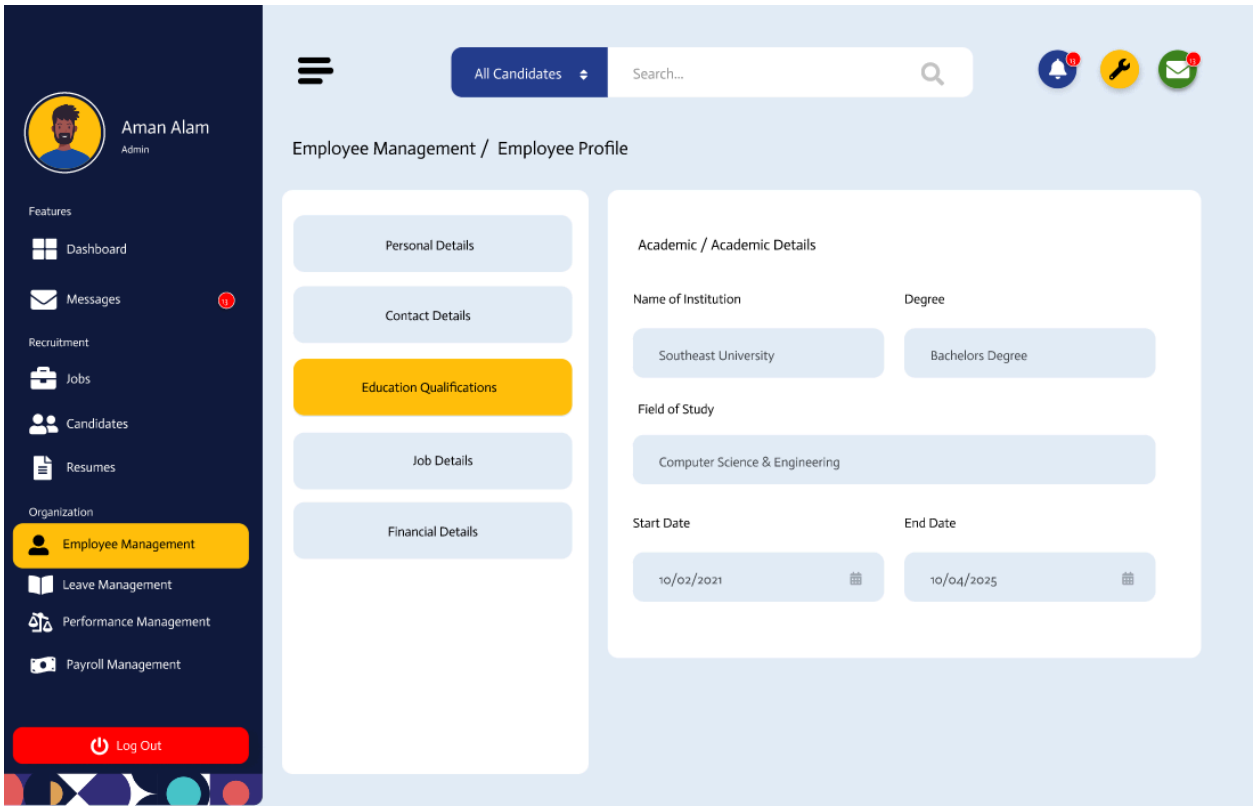
Admin Section:

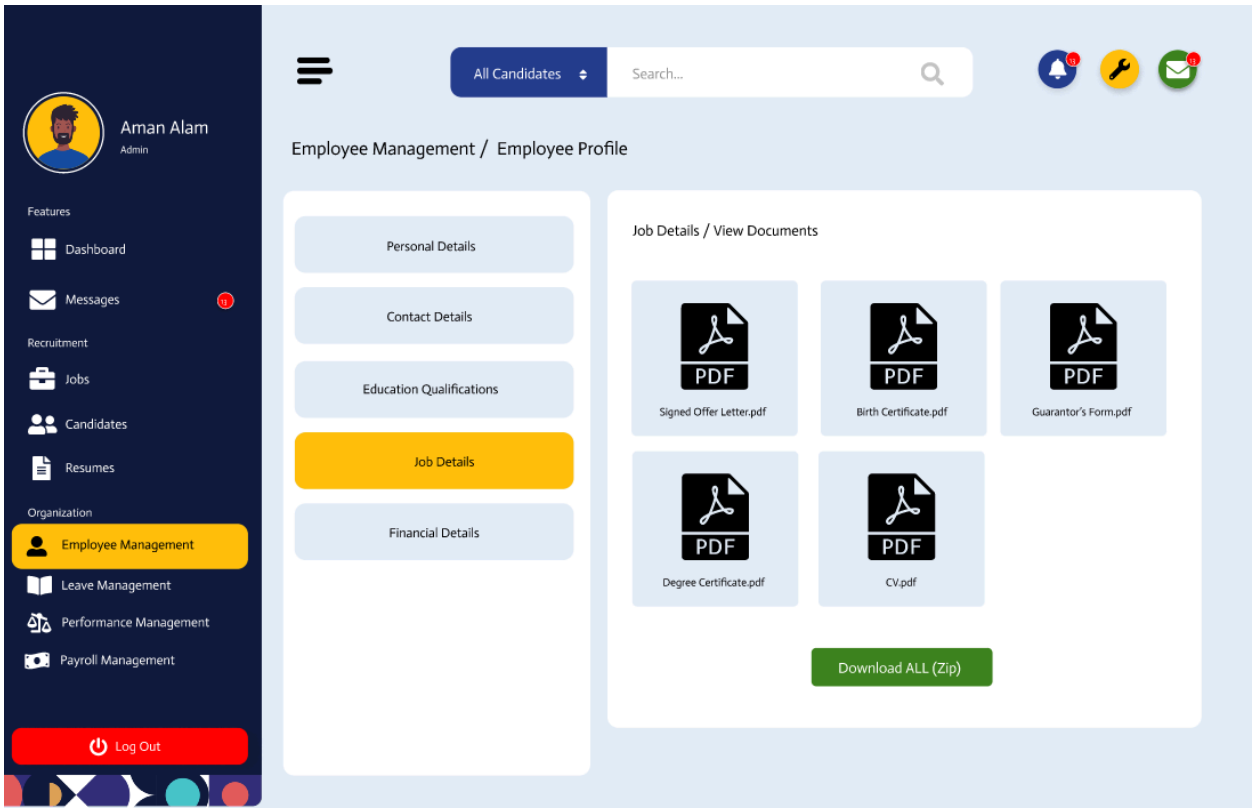
Employee Management

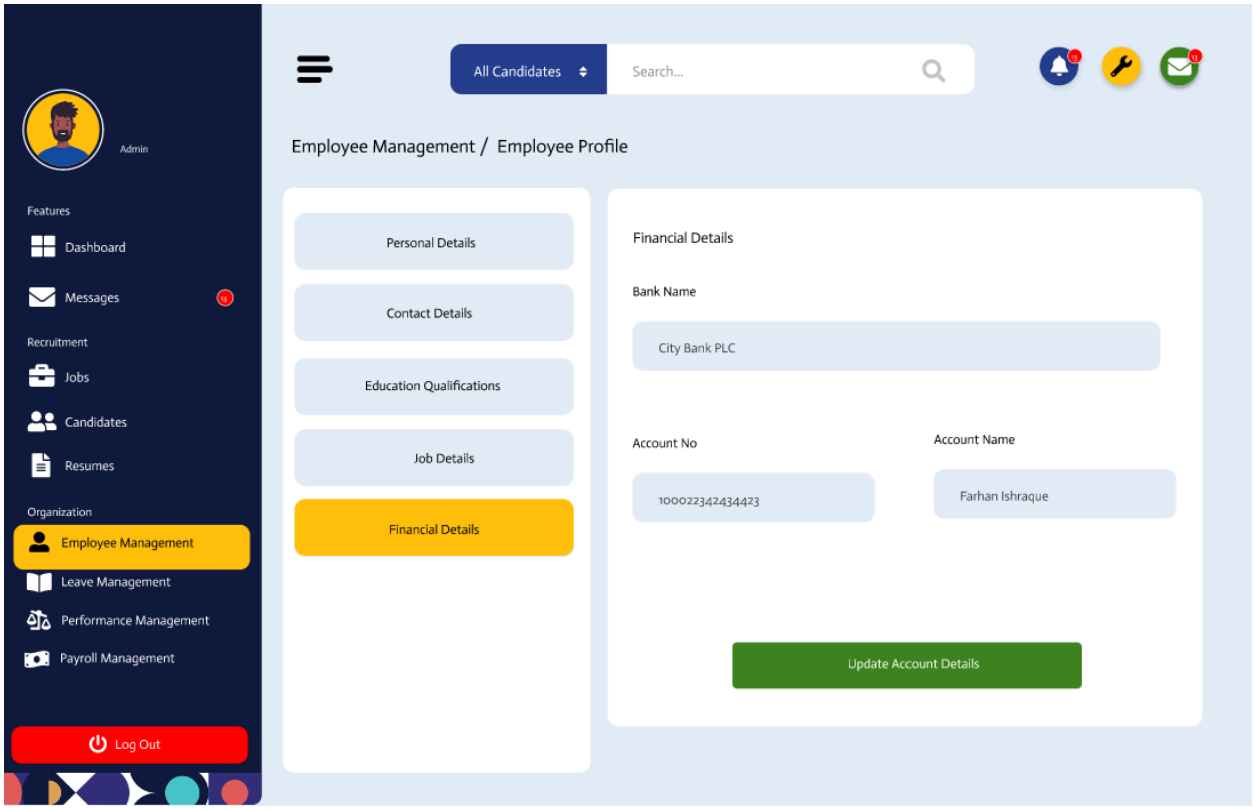



Aman Alam
Admin
Features
Dashboard
Messages
Recruitment
Jobs
Candidates
Resumes
Organization
Employee Management
Leave Management
Performance Management
Payroll Management
Log Out
All Candidates
Search...
Employee Management / Employee Profile
Personal Details
Contact Details
Educational Qualifications
Job Details
Financial Details
Edit
Employee Name
Farhan Ishraque
Department
Design & Marketing
Job Title
UI/UX Designer
Job Category
Full time


Aman Alam
Admin
Features
Dashboard
Messages
Recruitment
Jobs
Candidates
Resumes
Organization
Employee Management
Leave Management
Performance Management
Payroll Management
Log Out
All Candidates
Search...
Employee Management / Employee Profile
Personal Details
Contact Details
Education Qualifications
Job Details
Financial Details
Phone Number 1
01719840843
Phone Number 2
01435924683
E-mail Address
farhan34@gmail.com
City of residence
Dhaka
Residential Address
Maniknagar, Dhaka


Aman Alam
Admin
Features
Dashboard
Messages
Recruitment
Jobs
Candidates
Resumes
Organization
Employee Management
Leave Management
Performance Management
Payroll Management
Log Out
All Candidates
Search...
Employee Management / Employee Profile
Personal Details
Contact Details
Education Qualifications
Job Details
Financial Details
Academic / Academic Details
Name of Institution
Southeast University
Degree
Bachelors Degree
Field of Study
Computer Science & Engineering
Start Date
10/02/2021
End Date
10/04/2025


Aman Alam
Admin
Features
Dashboard
Messages
Recruitment
Jobs
Candidates
Resumes
Organization
Employee Management
Leave Management
Performance Management
Payroll Management
Log Out
All Candidates
Search...
Employee Management / Employee Profile
Personal Details
Contact Details
Education Qualifications
Job Details
Financial Details
Job Details / View Documents
PDF
Signed Offer Letter.pdf
PDF
Birth Certificate.pdf
PDF
Guarantor's Form.pdf
PDF
Degree Certificate.pdf
PDF
CV.pdf
Download ALL (Zip)


Admin
Features
Dashboard
Messages
Recruitment
Jobs
Candidates
Resumes
Organization
Employee Management
Leave Management
Performance Management
Payroll Management
Log Out
All Candidates
Search...
Employee Management / Employee Profile
Personal Details
Contact Details
Education Qualifications
Job Details
Financial Details
Financial Details
Bank Name
City Bank PLC
Account No
100023424344423
Account Name
Farhan Ishraque
Update Account Details

Testing:

Module	Test Cases			System Working Status	System Status	Remarks
Registration	**Email**	**Phone Number**	**Password**			
	x.com	01304948594	45678	Reg Successful	NOT OK	Invalid Email Format
	x@.com	00000004557	456789	Reg Not Successful	OK	Invalid Phone Number
	x@.com	01385749484	456789	Reg Successful	OK	All Okay
	Email		**Password**			
Log in	x@.com		456	Login Successful	NOT OK	Wrong Password
	x@.com		456789	Login Successful	OK	All Okay
User Dashboard	Update Profile			Takes to User Dashboard	OK	All Okay
	Apply Leave			Takes to Leave Application Dashboard	OK	All Okay
	Take Appraisal			Stays On User Dashboard	NOT OK	Do not Function

	View Payslip		Stays on User Dashboard	NOT OK	Do not Function
User: Leave Application	Annual Leave		Application Successful	OK	All Okay
	Sick Leave				
	Compassionate Leave				
Update Profile	Personal Details		Update Successful	OK	All Okay
	Contact Details				
	Email	x.com	Update Successful	NOT OK	Invalid Email Format
		x@.com	Update Successful	OK	All Okay
	Educational Qualifications		Update Successful	OK	All Okay
	Job Details				
	Financial Details				
Admin Dashboard	Employee Management		Takes to Employee Management Dashboard	OK	All Okay
	Leave Management		Takes to Leave Management Dashboard	OK	All Okay
	Performance Management		Stays on Admin Dashboard	NOT OK	Do not Function
	Payroll Management		Stays on Admin Dashboard	NOT OK	Do not Function

Admin: Leave Manage-Ment Dashboard	Leave Recall	Employees can be recalled	OK	All Okay
	Leave History	Leaves can be approved, declined and extended	OK	All Okay
Admin: Employee Manage-Ment Dashboard	Personal Details	Given information can be seen	OK	All Okay
	Contact Details			
	Educational Qualifications			
	Job Details	PDF documents can be downloaded	OK	All Okay
	Financial Details	Bank details can be seen and updated if neede	OK	All Okay

Conclusion:

The Employee Management System (EMS) provides an efficient and secure way to manage employee information, attendance, leaves and departmental operations. With its two-panel design, employees can easily access their profiles, view attendance, apply for leave and receive updates, while HR administrators can manage records, control access.

The system emphasizes security, scalability and user-friendliness, ensuring sensitive data is protected while remaining simple to use. Daily backups make the system reliable and adaptable for future organizational growth.

Overall, EMS improves efficiency, reduces errors and promotes transparency in workforce management. It saves time, enhances productivity and offers a structured approach to HR tasks, benefiting both employees and administrators.